



Overview of 365 Coffee Cafe

365 Coffee Cafe is a thriving coffee shop chain with three locations spread across New York City. Known for its premium coffee blends, cozy ambiance, and exceptional customer service, the cafe has become a favorite spot for locals and tourists alike. The business serves a wide range of products, including coffee, tea, pastries, sandwiches, and other snacks. With over 100,000 transactions recorded across its locations, the cafe has amassed a wealth of data that can be leveraged to optimize operations, improve customer experience, and drive revenue growth.

As part of their expansion and operational efficiency strategy, the franchise owners are keen on understanding customer purchase behavior, identifying trends, and uncovering opportunities to streamline operations. To achieve this, they have partnered with Power BI Certification students to analyze their transactional data and create a dynamic, interactive dashboard that provides actionable insights.

Project Overview:

The task assigned to the Power BI Certification students is to transform the provided dataset of over 100,000 transactions into a comprehensive and interactive dashboard. The dashboard should enable franchise owners to:

1. Analyze revenue trends over time.
2. Understand customer purchase patterns by day and hour.
3. Identify popular product categories and top-selling items.
4. Make data-driven decisions to optimize operations and increase profitability.

The dashboard should be visually appealing, easy to navigate, and provide actionable insights that can be used to improve business performance.

Requirements:

The project requires students to analyze the dataset and derive insights for the following key areas:

1. **Revenue by Month**
 - Analyze the total revenue generated each month across all locations.
 - Identify seasonal trends or months with the highest and lowest revenue.
 - Visualize the data using a line chart or bar chart to show monthly revenue trends.



2. Number of Transactions by Weekday

- Determine the busiest and least busy days of the week.
- Identify patterns in customer visits, such as higher traffic on weekends or weekdays.
- Use a bar chart or column chart to display the number of transactions by weekday.

3. Number of Transactions by Hours

- Analyze the distribution of transactions throughout the day.
- Identify peak hours and low-traffic hours to optimize staffing and inventory management.
- Visualize the data using a line chart or histogram to show transaction trends by hour.

4. Number of Transactions by Category

- Categorize transactions by product type (e.g., coffee, tea, pastries, sandwiches).
- Identify the most and least popular product categories.
- Use a pie chart or bar chart to display the distribution of transactions by category.

5. Top 15 Product Types and Revenue by Number of Transactions

- Identify the top 15 best-selling products based on the number of transactions.
- Analyze the revenue generated by these top products.
- Use a combination chart (e.g., bar chart for transactions and line chart for revenue) to visualize the data.

Expected Outcome

The final delivery is an interactive Power BI dashboard that provides franchise owners with a clear understanding of:

- Revenue trends over time.
- Customer purchase behavior by day, hour, and product category.
- Top-performing products and categories.

The dashboard should empower franchise owners to make informed decisions, such as:

- Adjusting staffing levels during peak hours.



- Optimizing inventory for high-demand products.
- Launching targeted promotions for underperforming categories or time slots.

Focus on building the BI tool with these specific insights in mind. The data you're working with should give us some solid starting points, but feel free to dig deeper where needed to provide actionable insights. Let me know if you need anything specific to move forward.

Kind regards,

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Manager, Business Intelligence

365 Coffee Cafe Inc.